American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Software Engineering

AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation

Submitted To

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Project Software Requirements Specification (SRS)

Submitted By

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Software Requirements Specification (SRS)

Project Title

AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation

Objective

Enhance the AIUB Portal (https://portal.aiub.edu/) to streamline student services, improve accessibility, and integrate smart technologies for course registration, navigation, notices, gymnasium services, complaints, resources, printing, faculty information, file sharing, and class scheduling.

1. Feasibility & Scope

- Technical Feasibility: Utilizes modern frameworks (React, Node.js), AR SDKs (ARKit, ARCore), and cloud infrastructure (AWS) for scalability and performance.
- Operational Feasibility: User-friendly interface accessible to students, faculty, and admins, supporting diverse user needs.
- Financial Feasibility: Modular development aligns with budget constraints; potential monetization via premium features.
- Scope: Enhance course registration, lost and found, smart location navigation, event notices, gymnasium registration, anonymous complaints, course resources, global printing, faculty directory, file sharing, and class scheduling.
- Scalability: The architecture must support increased user load, particularly during peak times such as course registration periods.

2. Functional Requirements

2.I. CGPA & Demand-Based Course Registration System

1. The system shall allow students to register for courses based on **batch priority** and **CGPA slots** to ensure fair access to high-demand courses.
2. CGPA-based registration slots shall be defined as:
    - **High CGPA (3.50 – 4.00):** First slot
    - **Mid CGPA (3.00 – 3.49):** Second slot
    - **Low CGPA (< 3.00):** Final slot
3. The system shall open registration **batch-by-batch** with fixed time windows to prevent server overload.
4. Students must submit a **pre-registration form** before midterm, including:
    - Current CGPA
    - Completed credits
    - Courses desired for next semester
    - Credit limit preference
5. The system shall automatically analyze **student demand** and open course sections accordingly.
6. The system shall **detect and prevent scheduling conflicts** in real-time during registration.

7. The system shall implement a **waitlist mechanism**:
   - If a section is full, students can join a waitlist.
   - When a seat is available, the system automatically allocates it based on waitlist priority.
8. The system shall allow students to submit **course priority requests** if they cannot register for required subjects.
9. Advisors shall be able to **approve and adjust registrations online**, removing the need for physical queues.
10. All registration activities shall be **time-to-time** for audit and dispute resolution.

Priority Level: High
Precondition: Students must be logged into the system.

2.II. Lost and Found Submission

1. Users shall be able to submit reports of lost or found items with a detailed description and images.
2. The submission shall be stored in a searchable database with filters (date, category, location).
3. Admin approval is required before the post becomes publicly visible.
4. Users shall be able to update or remove their submissions.

Priority Level: High
Precondition: Valid access credentials.

2.III. Smart Location System (AI + AR Navigation)

1. The system shall provide a **real-time interactive map** of AIUB's campus including **all buildings, floors, and room numbers**.
2. The location system shall integrate:
   - **Indoor Positioning System (IPS)** using Wi-Fi RTT, BLE beacons, or UWB.
   - **AI-based image recognition** to identify building markers and signs via camera.
   - **Augmented Reality (AR)** overlays to show directional arrows in the live camera feed.
3. Users shall be able to search for:
   - Classrooms (e.g., "C-324")
   - Faculty offices
   - Labs, library, gym, event halls
4. The system shall display **step-by-step guidance**:
   - Turn-by-turn navigation inside buildings
   - "Up/Down" indicators for stairs or elevators
5. The AR overlay shall:
   - Show arrows for left/right turns
   - Indicate distance to the next step
   - Provide text prompts ("Go Upstairs", "Turn Left")
6. **Voice guidance** shall be available for navigation.
7. The system should have an **accessibility mode** for wheelchair-friendly routes.

8. The map database shall be **regularly updated** to reflect room relocations or building changes.
9. The system shall work in **offline mode** inside campus after the initial map is caught.

Priority Level: Medium
Precondition: Users must access the map interface; camera and location permissions must be granted.

2.IV. Event and Notice Display

1. The system shall display **categorized notices** and **events** for previous, current, and upcoming dates.
2. Users shall be able to filter notices by **type**, **date**, and **audience**.
3. Students can register for **competitions, seminars, workshops** directly from the notice board.
4. Only authorized admins shall publish or modify notices.

Priority Level: Medium
Precondition: Admin publishing access.

2.V. Gymnasium Registration & Payment

1. Students can register for gym memberships online.
2. Payment tracking shall be integrated with the system.
3. Payment confirmation shall require admin verification.
4. Students can view membership history and renewal dates.

Priority Level: Medium
Precondition: Logged-in student account.

2.VI. Anonymous Complaint System

1. Students can submit complaints anonymously without revealing their identity.
2. Complaint data shall be encrypted and only accessible to higher authorities.
3. The system allow status tracking for submitted complaints without exposing identity.

Priority Level: Medium
Precondition: Anonymous submission form enabled.

2.VII. Course Book Reference & Online Resources

1. Authorities shall maintain a section with **reference books** for all courses.
2. Each entry shall include book details and links to **soft copies or online resources** (if legally permissible).
3. Students can search and download available resources.

Priority Level: Medium
Precondition: Authorities upload book details and resources.

2.VIII. Global Printing System

1. The system shall allow students to **upload documents remotely** through the portal from any device.
2. Students can select:
   - Printer location (any AIUB campus building)
   - Paper size & type
   - Number of copies
   - Color or grayscale printing
3. Documents shall be stored **securely in a print queue** until the user authenticates at the printer.
4. Printing shall require **student ID card scanning** or secure PIN entry at the printer.
5. The system shall integrate with a **cloud print management tool** similar to "SafeQ" to:
   - Enable global printing from any location
   - Prevent dependency on specific AIUB PCs
6. The system should notify students when documents are **ready for pickup**.
7. Payment options shall include:
   - Deduction from student account balance
   - Online banking payment
8. Admins shall have access to **print usage reports** for tracking and auditing.

Priority Level: High
Precondition: Students must be logged in; license and system updates must be applied.

2.IX. Faculty Information Directory

1. Searchable profiles of faculty members with department, contact info, office location, and teaching courses.
2. Faculty profiles shall also list research areas and publications.
3. Directory updates shall be managed via the admin panel.

Priority Level: Medium
Precondition: Directory database initialized.

2.X. AIUB File Sharing System

1. Users can send and receive files within the AIUB network through the portal.
2. The system shall support temporary **secure links** for file sharing.
3. Files shall auto-delete after an expiry period to save storage and ensure privacy.

Priority Level: Low
Precondition: Authenticated AIUB network user

2.XI. Class Routine & Upcoming Registration Course Generator

1. The system shall detect schedule conflicts during registration.
2. It should suggest eligible courses for the upcoming semester based on prerequisites.
3. Students can generate a **personalized routine** with preferred time slots.

Priority Level: Medium
Precondition: Logged-in student account.

3. Non-Functional Requirements

- **Performance**: Respond to user actions within 2 seconds.
- **Security**: Use HTTPS, role-based access control, and encryption for anonymous complaints.
- **Usability**: Mobile-friendly, intuitive, accessible interface supporting disabilities.
- **Reliability**: Ensure high availability with minimal downtime; perform daily backups.
- **Scalability**: Support increased user load during peak times (e.g., registration).

4. Implementation Phases

Phase 1: Research & Development (3 months):

- Conduct literature review on AR navigation, AI-driven registration, and cloud printing systems.
- Select technologies (React, Node.js, MySQL, ARKit, ARCore, Navigine SDK, AWS).
- Design system architecture and defining datasets (e.g., course demand, campus map).
- Create initial prototypes for critical modules (e.g., AR navigation, registration form).

Phase 2: Iterative Development & Testing (6 months):

- Develop modules in 2–3-week sprints: course registration, smart location system, global printing, etc.
- Implement AR navigation with BLE beacons and QR code markers; test accuracy in one AIUB building.
- Build and test online payment integration for printing and gymnasium services.
- Train AI models for course demand analysis and route optimization; validate with test data.
- Conduct unit and integration testing for each sprint; address bugs and performance issues.

Phase 3: Pilot Testing (4 months):

- Deploy initial modules (e.g., registration, printing, navigation) in a controlled environment (e.g., one department).
- Collect feedback from students, faculty, and admins on usability (e.g., AR directions, payment flow).
- Refine features based on feedback; optimize performance (e.g., response time <2 seconds).
- Test scalability during simulated peak registration periods.

Phase 4: Full Deployment & Optimization (3 months):

- Roll out all modules across AIUB campus via the portal (https://portal.aiub.edu/).
- Monitor system performance during high-traffic events (e.g., registration, event notices).
- Implement daily backups and security updates (HTTPS, RBAC, encryption).
- Address user-reported issues and deploy updates iteratively.
- Document system maintenance procedures for admins.

5. Tools & Technologies

- **Frontend**: React.js
- **Backend**: Node.js
- **Database**: MySQL or SQL Server
- **AR Navigation**: ARKit, ARCore, Navigine SDK
- **AI**: TensorFlow, Scikit-learn
- **Cloud**: AWS
- **Version Control**: Git
- **Beacons**: BLE for indoor positioning
- **3D Mapping**: Unity

6. Evaluation Metrics

- Accuracy of course demand analysis and registration allocation.
- AR navigation precision and response time (<2 seconds).
- User engagement (daily active users, event registrations).
- System uptime and reliability (>99%).
- Feedback quality from pilot testing.

7. Future Enhancements

- IoT integration for real-time facility tracking (e.g., classroom occupancy).
- AI-driven predictive analytics for course demand forecasting.
- Expanded global printing with third-party integrations.
- Enhanced AR features for virtual campus tours.

Conclusion

The enhanced AIUB Portal will integrate smart digital solutions to improve campus navigation, registration, and engagement. Using Agile methodology, modular development, and advanced technologies like AR and AI, the system will deliver a scalable, user-centric platform for AIUB's community.